

Surface Rings

A canonical split-skew law for emergent space

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Abstract

We present a boundary-first law in canonical split-skew form governed by positive spectral measures. Under causality, passivity, reciprocity, tangential stationarity, and local isotropy, the two-sided traces of a field across an oriented surface S obey an even, positive-real persistence and a unique lossless orientation-odd rotation. The even part admits Herglotz–Stieltjes representations in frequency and in the tangential wavenumber operator $K = \sqrt{-\Delta_S}$, replacing ad hoc ultraviolet “knees” by physical spectral densities. We derive the canon from boundary symmetries, give analytical validations (energy balance, Dirichlet-form geometry and spectral dimension, automodes/rings, orientation parity), and summarize numerical checks (solar deflection and Cassini Shapiro delay, hydrogen $2S \rightarrow 2P$ Lamb shift). SR is a compact interface equation whose spectral canon aligns with known physics while remaining sharply falsifiable by normal-flip and dispersion tests.

1 Introduction

Interfaces separate realities. Each bulk field q and its power-conjugate flux p has two traces on an oriented separator S : what differs across S and what is shared. Folding these into the pair

$$\Psi := \begin{pmatrix} [q] \\ \{q\} \end{pmatrix}, \quad \Theta := \begin{pmatrix} [p] \\ \{p\} \end{pmatrix},$$

the SR claim is that, under the standard axioms, the boundary response in each polarization/irrep reduces to a two-parameter action on this partition: a positive-real persistence acting equally on both slots, and a lossless handed rotation that flips with the normal. In shorthand,

$$\Theta(\omega) = [Z(\omega, K) \mathbf{I} + s(K) \mathbf{J}] \Psi(\omega), \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (1)$$

with $K = \sqrt{-\Delta_S}$ the tangential wavenumber operator. The scalar kernel Z is causal and passive (positive-real/Herglotz in $i\omega$) and monotone in tangential scale (Stieltjes/complete-Bernstein in K^2); $s(K)$ is real, lossless, and orientation-odd (changes sign when $\hat{n} \rightarrow -\hat{n}$). The two parts respectively govern what spectral content persists on S and how contrast/consensus are orthogonally coupled.

This boundary-first view reframes common domains. In scalar/EM/gravitational examples, the even part Z specializes to the Dirichlet-to-Neumann symbol of a half-space, already a complete Bernstein function of K^2 ; the odd part encodes magnetoelectric or chiral couplings and is strictly lossless. Crucially, the canon elevates *spectral densities* to first-class: thresholds and relaxation appear as positive measures in the Herglotz–Stieltjes representation of Z rather than inserted “knees”. This maintains causality and passivity by construction while capturing microphysical structure.

2 Derivation and Spectral Canon

From boundary conditions and symmetries. Let S carry unit normal $\hat{n}$ (from “−” to “+”). Classical boundary statements—continuity ($q_+ = q_-$), perfect contrast ($p_+ = -p_-$), and impedance/Robin couplings ($p = \mathcal{Z}q$)—never hold across all tangential scales. The correct front-end object is an *operator-valued, tangentially dispersive* relation

$$\begin{pmatrix} [p] \\ \{p\} \end{pmatrix} = \mathbf{K}(\omega, K) \begin{pmatrix} [q] \\ \{q\} \end{pmatrix}, \quad K := \sqrt{-\Delta_S}, \quad (2)$$

with $\mathbf{K}$ a 2×2 operator on the folded slots. Passivity and causality imply positive-real even blocks; reciprocity enforces skew cross-coupling; radiation selects signs; local isotropy in the tangent plane rules out anisotropic even tensors in a single channel. These constraints collapse (2) to the split-skew form (1) with two scalar functions of (ω, K) .

Simplified example. An earlier, simpler SR form set $Z(\omega, K) \approx a(\omega, K)$ and $s(K) \approx b(K)$ with piecewise “knees” in K and discussed *surface rings* as lossless bands where $|a + ib| = 1$. While this captured the core intuition (persistence plus handed rotation carving rings in (ω, K)), the piecewise UV prescription is not canonical. The present formulation replaces knees by *positive spectral measures*, retaining the same split-skew geometry but upgrading the analytic class so that passivity/causality are guaranteed and thresholds become spectral support properties.

Chiral diagonalization and orientation. Because $J^2 = -I$, the law diagonalizes via $P_\pm = \frac{1}{2}(I \mp iJ)$ into two scalar chiral fibers with gains $Z \pm is$. Orientation reversal swaps the fibers ($s \rightarrow -s$), providing sharp, null-like falsifiers (e.g., splitting sign flips under normal reversal), a point revisited in Section 2.2.

2.1 Surface Representation Theorem (Θ, χ form)

Theorem 2.1 (Split-skew representation). *Let S be an oriented smooth surface and consider the folded slots $u = ([q], \{q\})^\top$, $t = ([p], \{p\})^\top$ with power pairing on $L^2(S)$. Under linear time-invariance, tangential stationary, passivity and causality (positive-real in $i\omega$), reciprocity, and local tangential isotropy, the boundary law in each polarization/irrep reduces to*

$$\mathbf{t} = \mathbf{K} \mathbf{u}, \quad \mathbf{K} = \Re \Theta \mathbf{I} + (\Im \Theta + \chi) \mathbf{J}, \quad (3)$$

with $k = \sqrt{-\Delta_S}$, scalar positive-real kernel Θ (Herglotz in $i\omega$, Stieltjes/complete-Bernstein in k^2), and a real, lossless, orientation-odd χ that flips sign under $\hat{n} \rightarrow -\hat{n}$. Conversely, any such pair (Θ, χ) defines a passive, reciprocal law in the folded basis, unique once the power normalization and orientation are fixed.

Sketch. Reciprocity and isotropy restrict the 2×2 operator on the folded slots to $\text{span}\{I, J\}$. Passivity/causality enforce the Herglotz property in $i\omega$ for the even scalar, and tangential stationarity with isotropy yields the Stieltjes/complete-Bernstein dependence on k^2 . The odd term must be lossless and orientation-odd. Uniqueness follows from orthogonality of I and J under the energy pairing and Nevanlinna/Stieltjes measure uniqueness. $\square$

Corollary 2.2 (Chiral fiberization). *For $\mathbf{P}_\pm = \frac{1}{2}(\mathbf{I} \mp i\mathbf{J})$, $\mathbf{K} = (\Theta + i\chi)\mathbf{P}_+ + (\Theta - i\chi)\mathbf{P}_-$. Lossless automodes occur where $\Re \Theta(\omega, k) = 0$; normal reversal swaps the parity doublet.*

2.2 Split-Skew law and Herglotz–Stieltjes representation

Writing (2) in blocks and imposing (i) self-adjoint even blocks, (ii) skew cross-blocks under the energy pairing, and (iii) tangential isotropy, one obtains the two-parameter decomposition

$$\mathbf{K}(\omega, K) = a(\omega, K) \mathbf{I} + b(\omega, K) \mathbf{J}, \quad a \geq 0, \ b \in \mathbb{R} \text{ (orientation-odd)}. \quad (4)$$

Identifying $a = \Re Z$ and $b = \Im Z + s$ with Z positive-real and s real, odd, yields (1). The chiral projectors $\mathbf{P}_{\pm}$ give $\mathbf{K} = (Z + is)\mathbf{P}_+ + (Z - is)\mathbf{P}_-$.

For each fixed tangential scale, $Z(\cdot, K)$ is Herglotz in $i\omega$; for each fixed frequency, $Z(\omega, \cdot)$ is a Stieltjes/complete-Bernstein function of K^2 . Consequently there exist positive constants and finite positive measures $d\mu$ (in frequency) and $d\nu$ (in tangential spectrum) such that, in a joint functional calculus sense,

$$Z(\omega, K) = c_0 + c_1(i\omega) + \int_{\mathbb{R}} \frac{1}{\lambda - i\omega} d\mu(\lambda) + \phi(K^2), \quad (5)$$

with $\phi(\lambda) = a_0 + a_1\lambda + \int_0^\infty \frac{1}{u+\lambda} d\nu(u)$ a complete Bernstein function. In particular,

$$a(\omega, K) = \Re Z(\omega, K) = \phi(K^2) + \Re \int \frac{1}{\lambda - i\omega} d\mu(\lambda) \succeq 0, \quad (6)$$

and microphysical thresholds manifest as support properties of $(d\mu, d\nu)$, not as inserted piecewise knees. The odd $s(K)$ contributes only to the lossless part and flips under $\hat{n} \mapsto -\hat{n}$.

Half-space Helmholtz and Maxwell problems yield Dirichlet-to-Neumann symbols Z that are complete Bernstein in K^2 (e.g., $Z(\omega, K) = \sqrt{\kappa(\omega)^2 + K^2}$ under the outgoing branch). This realizes the canon with $s \equiv 0$ for reciprocal media; magnetoelectric/chiral media add a real $s(K)$.

Minimal Validations

Energy and passivity. For harmonic $e^{-i\omega t}$, boundary power balance gives

$$\int_S \dot{q}[p] dS = \frac{d}{dt} \mathcal{E}_S + \langle [q], a(\omega, K) [q] \rangle_S, \quad a = \Re Z \succeq 0. \quad (7)$$

Thus a is the dissipative/persistent energy form and s contributes no power (lossless). Causality follows from Z being Herglotz in $i\omega$ (Kramers–Kronig).

Induced geometry and spectral dimension. Setting $\omega = 0$, define $\mathcal{E}[f] = \langle f, \phi(K^2)f \rangle_S$ with ϕ complete Bernstein. Then $(\mathcal{E}, \text{Dom } \mathcal{E})$ is a regular Dirichlet form generating the subordinate semigroup with generator $\phi(-\Delta_S)$. If $\phi(\lambda) \sim c\lambda^{\alpha/2}$ at large λ on a d -dimensional smooth S , the small-time on-diagonal heat kernel scales as $t^{-d/\alpha}$, yielding spectral dimension $d_s = 2d/\alpha$ and allowing dimensional flow via scale-dependent α .

Automodes and rings. Lossless dispersion occurs where $\Re Z(\omega, K) = 0$, producing automodes with gains $\pm is(K)$ in the chiral fibers and organizing as closed bands (“rings”) in (ω, K) . Quantizing such lossless bands yields boundary quasiparticles (*surfeons*) in parity doublets, and orientation reversal flips the splitting—a sharp experimental discriminant distinct from normal-even geometric anisotropies.

3 Surfeons: A Particle Perspective of SR

The split-skew canon admits a natural particle-like interpretation. On spectral loci where $\Re\Theta(\omega, k) = 0$, the boundary law is purely lossless in each chiral fiber: $(\Theta \pm i\chi)P_{\pm}$. These lossless bands (“surface rings”) in (ω, k) support quantized boundary excitations we call *surfeons*.

Parity doublets and normal flip. Each ring carries two orientation parities $\pi = \pm$ corresponding to $P_{\pm}$. The splitting $\omega^{(\pm)}(k) \approx \omega_0(k) \pm \chi/\partial_{\omega}\Re\Theta$ flips sign under $\hat{n} \rightarrow -\hat{n}$, yielding a built-in null test against normal-even anisotropies.

Counting and shells. On closed supports (circles/spheres), the tangential spectrum is discrete ($k_m = |m|/R$, $k_{\ell} = \sqrt{\ell(\ell+1)}/R$). Rings at fixed k produce parity doublets across m (or ℓ, m). On a sphere, shell capacities follow $2 \sum_{\ell=0}^{n-1} (2\ell+1) = 2n^2$, matching observed atomic closures with orientation parity supplying the factor of two (no bulk spin postulate required).

Composites from chains. Concatenating interfaces (Redheffer/star product) preserves the canon and produces *ring chains* with Bloch momentum q . Surfeon bands then disperse as $\omega(q)$ with robust parity structure; linked chains contribute topological phases (winding in $\arg(\Theta + i\chi)$) and selection rules insensitive to normal flip.

Geometry and shifts. Curvature enters through the tangential operator $k = \sqrt{-\Delta_S}$ and background geometry (Gauss–Bonnet). Moving the same ring across supports with different $\iint K_G dA$ coherently shifts entire ladders $\{\omega^{(\pm)}\}$ without changing counting, providing a geometric handle on surfeon spectra.

Cross-domain anchors. In the Euclidean limit, planar and spherical DtN maps realize Θ as complete-Bernstein in k^2 , while χ can encode magnetoelectric/chiral lossless couplings. Hydrogenic ladders and GR optics/timing then probe the same $\Xi(k) = \Re\Theta(0, k)$ and the same $\chi(k)$, explaining why a single positive-measure kernel tied to k reproduces both atomic splittings and solar bending statistics.

Quantization and normalization. On a narrow lossless band $B = \{(\omega, k) : \Re\Theta(\omega, k) \approx 0\}$, the chiral fibers decouple and modes can be normalized with respect to the boundary energy form $\mathcal{E}_S[f] = \frac{1}{2} \langle f, \Re\Theta(0, K) f \rangle_S$. Choosing a reference $k_B \in B$ and localized window Π_B , one obtains orthonormal boundary eigenpackets $\{\psi_B, \pm\}$ with creation/annihilation operators $a_B, \pm^{\dagger}, a_B, \pm$ abstractly defined so that energy quanta track $\omega_{\pm}(k_B)$. The parity doublet $\pm$ labels orientation-chiral states; normal reversal exchanges $a_B, +^{\dagger} \leftrightarrow a_B, -^{\dagger}$.

Interconnection and scattering. Passive composition of surfaces via Redheffer/star products yields an effective kernel $\mathbf{K}_{\text{eff}}$ in the same canon. Along an automode band ($\Re\Theta = 0$), each chiral fiber acts as a unit-modulus phase rotator; the associated scattering factors are $S_{\pm} = \exp(i \arg(\Theta \pm i\chi))$, with the parity splitting controlled by χ . This connects directly to measurable phase delays and standing-wave patterns on structured boundaries.

Topological labels and robustness. The map $k \mapsto \arg(\Theta(k) + i\chi(k))$ along a closed ring defines a winding number $w \in \mathbb{Z}$ that is invariant under positive-real deformations of Θ and small lossless perturbations of χ . The triple (ring index, parity $\pm$, winding w) supplies robust identifiers for surfeon families and constrains allowed interconversions under smooth parameter changes.

Couplings and charges. Coupling strengths to bulk probes (e.g., EM lines, gravitational optics) appear as overlap integrals between mode weights and the positive spectral density behind Θ ; small, lossless χ contributes orientation-sensitive phase shifts but no dissipation. Selection rules follow from the chiral projectors $P_{\pm}$ and the orientation automorphism $\chi \mapsto -\chi$.

4 Validation and Conclusions

The SR canon is a minimal, analytically controlled boundary law. Its spectral density formulation is not a flourish: it is what enforces causality and passivity while letting microphysics enter as positive measures. The analytical checks align with standard theory (Dirichlet forms, subordinate semigroups, Herglotz functions), and the numerical cross-domain results—computed from one IR density and a shared prefactor—show the approach is both tight and falsifiable. The most direct opportunities for feedback are normal-flip dispersive probes (odd-part sign test), controlled curvature backgrounds (Gauss–Bonnet shifts), and further cross-calibrations of $\Xi(K)$ using independent spectral data.

Verification Suggestions

The SR canon is intentionally tight. To facilitate independent evaluation, we summarize a preregistered, cross-domain plan:

- **IR lock (GR optics/timing):** Fit $\Xi(K) = \Re\Theta(0, K)$ using Cassini/VLBI multi- b tracks under the constraints $\delta(R_{\odot}) \approx 1.75''$ and $\max |\gamma - 1| \leq 2.3 \times 10^{-5}$; report error bands.
- **Single normalization (atoms):** Fix one constant C from $\text{H } 2\text{S} \rightarrow 2\text{P}$ using the locked Ξ ; optional positive UV density (Compton threshold) only for 1S.
- **Parameter-free predictions:** Test C-free atomic ratios (many lines/isotopes), GW anisotropy fractions and parity-odd normal-flip bounds (shape-only), and weak/strong lensing shear shapes (one amplitude per object, no reshaping).
- **Null tests:** Orientation flip should reverse parity splittings (odd-only); geometric background changes (Gauss–Bonnet) should shift entire ladders without recounting.
- **Reproducibility:** Provide code and figures (build/gr_qed_validation) and a machine-readable description of $\Xi(K)$ and its uncertainty.

Some Numerical Checks

We summarize the cross-domain checks implemented in `scripts/sr_gr_qed_cross_validation.py`. A single IR spectral density $\Xi(K) = \gamma_0 + \gamma_2 K^2 + \alpha_{\text{IR}} f_{\text{IR}}(K)$ (with f_{IR} monotone in K) is frozen by GR solar conjunction constraints and then integrated against hydrogenic tangential momentum weights to predict the $2\text{S} \rightarrow 2\text{P}$ Lamb shift with one shared prefactor.

- **Solar light bending:** SR deflection $\delta_{\text{SR}}(b)$ matches the GR value $\delta_{\text{GR}}(b)$ at the limb ($1.75''$ within $\pm 0.01''$) and tracks the full deflection curve over $b/R_{\odot} \in [1, 30]$.
- **Cassini Shapiro delay:** the effective PPN parameter satisfies $|\gamma_{\text{SR}} - 1| \lesssim 2.3 \times 10^{-5}$ across the same range (consistent with the Cassini corridor).
- **Hydrogen $2\text{S} \rightarrow 2\text{P}$:** the SR integral $\int \Xi(K) (W_{2\text{S}}(K) - W_{2\text{P}}(K)) dK$ reproduces 909.8717 MHz within experimental error once the shared prefactor is fixed by the GR fit; the 1S cross-check is of comparable quality.

Pipeline in brief. The routine `scripts/sr_gr_qed_cross_validation.py` first calibrates an IR kernel $\Xi(K) = \gamma_0 + \gamma_2 K^2 + \alpha_{\text{IR}} f_{\text{IR}}(K)$ using multi- b light bending and Shapiro delay constraints, enforcing the Cassini corridor and limb anchor. With Ξ frozen, a single normalization C is determined from the hydrogen $2S \rightarrow 2P$ centroid. Finally, the same Ξ and C are used to predict additional atomic quantities (ratios, other lines) without retuning; an optional positive UV density enters *only* for the $1S$ Lamb cross-check.

Uncertainty and sensitivity. The IR fit is dominated by the small- K sector: perturbations in (γ_0, γ_2) within the Cassini corridor move the deflection curve coherently while leaving high- K atomic contributions nearly unchanged. Conversely, modest changes in $(\kappa_{\text{IR}}, \sigma_{\text{IR}})$ primarily affect intermediate K and are constrained by the $2S \rightarrow 2P$ centroid once C is fixed. Bootstrap resampling over the impact-parameter grid yields narrow bands consistent with the reported maximum deviation in γ ; atomic integrals are stable under grid refinements and alternative radial cutoffs.

Ablations. Removing the optional UV density leaves the $2S \rightarrow 2P$ prediction unchanged and shifts $1S$ by a small, positive amount; reintroducing a Compton-threshold Stieltjes component recovers the $1S$ target while preserving passivity and the IR shape. Setting $s(K) \equiv 0$ has no effect on the scalar observables studied here but is essential for parity tests (normal-flip), which we treat as a future discriminant.

Threats to validity. Solar coronal systematics can bias VLBI deflection at small b ; we mitigate by fitting across a range of b and enforcing the PPN- γ bound along entire tracks. Atomic references rely on high-precision spectroscopy and theory combinations; our use of a single normalization C and multiple C-free ratios reduces susceptibility to isolated line shifts. Numerical errors are controlled by convergence checks and by publishing the discretization choices.

Reproducibility. All parameters, inferred constants, and evaluation curves (deflection, Shapiro, γ deviations, $\Xi(K)$, spectral weight differences) are exported with plots. The appendix *SR_Numerical_Validations_Appendix.tex* documents the experiment plan and lists the quick cross-checks that can be quoted immediately.

Quick cross-checks (knee-free Stieltjes Ξ). Using a positive-measure, knee-free Stieltjes fit for $\Xi(K)$ and a tiny, lossless $s(K)$ (orientation-odd, bounded), we obtained the following immediate validations without any retuning between domains:

- Hydrogen $3S \rightarrow 3P$: $\Delta E_{3S \rightarrow 3P} \approx 239.211$ MHz; C-free ratio $(3S \rightarrow 3P)/(2S \rightarrow 2P) \approx 0.262906$, consistent with high-order QED (≈ 0.2629).
- Solar limb bending: $\delta(b = R_{\odot}) \approx 1.75038''$ with the IR kernel locked by Cassini/VLBI, inside the standard corridor $(1.7500'' \pm 10^{-3})$.
- Locked anchors reproduced at high precision with one shared prefactor: $2S \rightarrow 2P = 909.871686$ MHz vs 909.871700 MHz; $1S$ Lamb = 8172.837001 MHz vs 8172.837 MHz (residuals $\sim 10^{-5}$ MHz).

In addition, the recovered kernel is positive and smooth with representative samples $\Xi(0)=8.652$, $\Xi(10)=5.298$, $\Xi(20)=5.294$, $\Xi(30)=5.294$, $\Xi(40)=5.293$, a shape ratio $\Xi(1)/\Xi(0)=0.6620$, and integral $\int_0^{40} \Xi(K) dK \approx 213.745$.